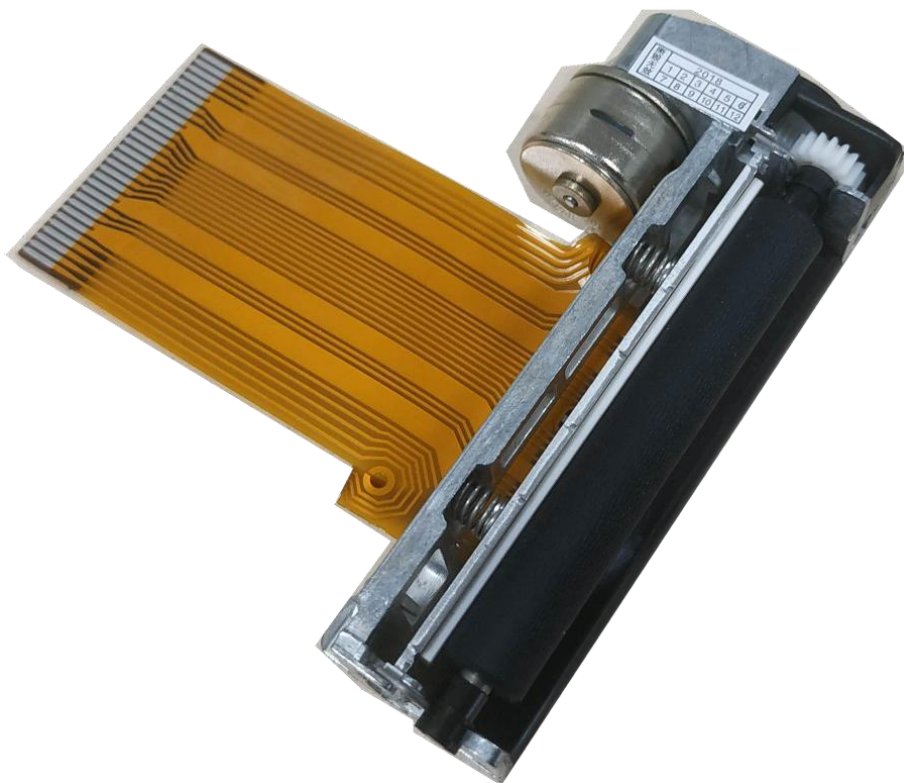


TP-486F-101
Thermal Printer Mechanism Specification
Compatible FUJI: FTP-628MCL101



Revision record

[illegible]

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1 Introduction

1.1 TP-486F thermal printer mechanism series

TP-486F thermal printer mechanism has the characteristics of small size, wide working voltage and high efficiency. The unique movement design makes the TP-486F easy to load paper and high reliability.

1.2 TP-486F feature

- ◆ Easy loading paper
- ◆ Small size
- ◆ Light weight
- ◆ Fast print speed (max print speed 90mm/s)
- ◆ Wide working voltage
- ◆ High DPI (8 dot//mm,203DPI)
- ◆ Long printing life (>50km)
- ◆ Low noise

1.3 Remark

This manual describes the electrical and mechanical characteristics of the TP-486F. For designers' reference.

The company reserves the right to change this manual. If you need the latest version of this manual, you can consult with us.

For the design of this manual, we do not guarantee the reliability of its work.

2 . Specifications

Item	specification
Printing method	Thermal dot printing
Number of dots	384 dots/line
Resolution	8dot/mm/(203dpi)
Valid Printing width	48mm
Paper width (mm)	58
Paper feed pitch	0.0625mm
Dimension(mm)	70.2x33x15.3
Weight (g)	42g
Head temperature	By NTC thermistor 30K Ω (25°C)
Paper end sensor	By photo interrupter
For printer head (V)	4.2~9.5
For logic (V)	2.7~5.25
Operating temperature (°C)	0~50 (no condensation)
Operating humidity (RH)	20%~85% (no condensation)
Storage temperature (°C)	-25~70 (no condensation)
Storage humidity (RH)	5%~95% (no condensation)

3 Thermal head specifications

3.1 Outlines

Item	specification
Number of Heat Elements	384dots/line
Dot density	8 dots/mm(0.125 mm pitch)
Dot size	0.11mm x 0.10mm
Paper feed pitch	0.0625mm
Valid printing width	48mm
Average Resistance Value	176 Ω $\pm$ 4%
Operating voltage	4.2~9.5V
Pulse life	10 ⁸
Mechanical life	50km

Life test conditions: 25 °C, the heat ing time ratio is not more than 12.5%.

3.2 Maximum Conditions

Item	Symbol	Maximum Reference	Remark
Supply Energy	E _o max	0.2mJ/dot	1.25ms/line
Supply Voltage	V _H	9.5V	Voltage across the cable
Logic voltage	V _{dd}	7V	Including peak voltage
Temperature	T _a	-30℃～50℃	>5℃
Humidity		10%～90%RH	no condensation
Substrate Temperature	T _s	Keeping 65℃、30 minutes	When the temperature reaches 80℃, must stop printing until the temperature drops to 60℃
		MAX 80℃	

3.3 Calculation formula

Supply Energy:

$$E_o = I_o^2 R t_s = \frac{VH^2 \times R_{ave} \times t_s}{(R_{com} \times N + R_{ave} + R_{IC} + R_{lead})^2}$$

Remark:

RIC 9Ω

ts time

VH Voltage

R_{ave} Average Resistance Value 176Ω

N Heating dots at the same time 64 dots

R_{co} Common Terminal Resistance Value 0.05Ω

R_{lea} connector line resistance

3.4 Reference data

Item		Symbol	Reference working data	Remark
Supply Energy		E _o	0.27W/dot	$R_{ave}=176\Omega$
Supply Voltage		V _H	7.2V	
Speed			1.25ms/line	
Supply Energy	5℃	E _o (ts)	0.16mJ/dot(0.6ms)	V _{dd} =5V 64dots heating dots in the same time
	25℃		0.13mJ/dot(0.49ms)	
	45℃		0.11mJ/dot(0.41ms)	
Current		I _o	37.5mA/dot	

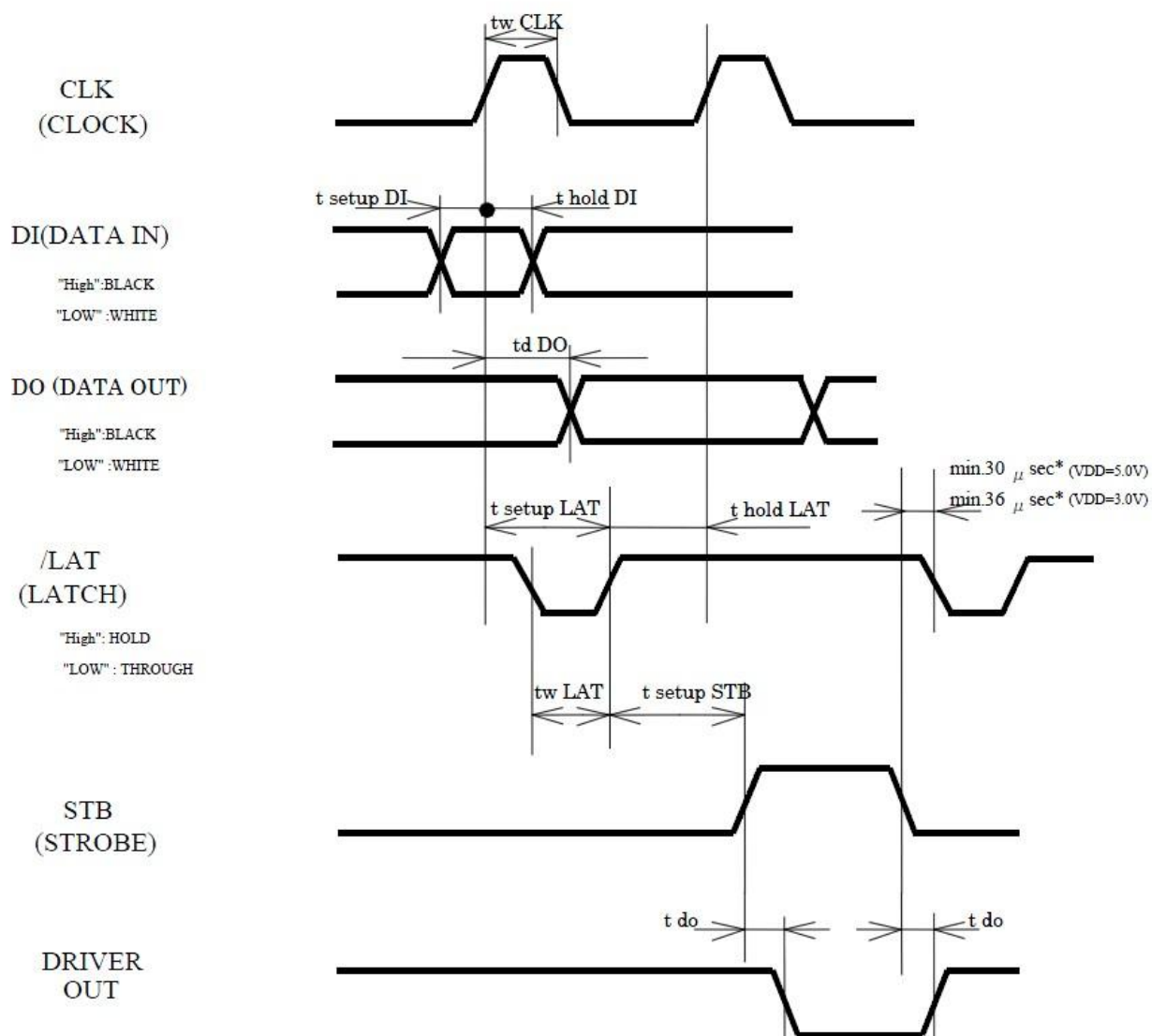
3.5 Electrical data 1 (V_{DD}=5V)

Item		Symbol	Min	Typical	Max	Unit	Condition
Print voltage		V _H			9.5	V	
Logic voltage		V _{DD}	4.75	5.00	5.25	V	
Logic current		I _{DD}			48	mA	f _{DI} =f _{CLK} /2 F _{clk} =10MHz
Input voltage	High	V _{IH}	0.8V _{DD}		V _{DD}	V	STB, DI, LAT, CLK
	Low	V _{IL}	0		0.2 V _{DD}	V	
Data input current	High	I _{IH} DI			0.5	μA	V _{IH} =5V V _{IL} =0V
	Low	I _{IL} DI			-0.5		
STB input current	High	I _{IH} STB			55		
	Low	I _{IL} STB			-0.5		
Clock input current	High	I _{IH} CLK			1.0		
	Low	I _{IL} CLK			-1.0		
Latch input current	High	I _{IH} LAT			1.0		
	Low	I _{IL} LAT			-1.0		
Output voltage	High	V _{DOH}	4.25			V	V _{DD} =5V, I _{OH} =-0.5mA
	Low	V _{DOL}			0.75	V	V _{DD} =5V, I _{OL} =0.5mA
Clock frequency		f CLK			10	MHz	Review Fig.3
Clock width		t _w CLK	30			ns	
Data setup time		t _{setup} DI	30			ns	
Data retention time		t _{hold} DI	30			ns	
Data output delay time		t _d DO			80	ns	
Latch pulse width		t _w LAT	40			ns	
Latch setup time		t _{setup} LAT	60			ns	
Latch retention time		t _{hold} LAT	20			ns	
STB setup time		t _{setup} STB	300			ns	
Heating to drive output delay		T _{do}			30	μs	

3.6 Electrical data 2(V_{DD}=3.0V)

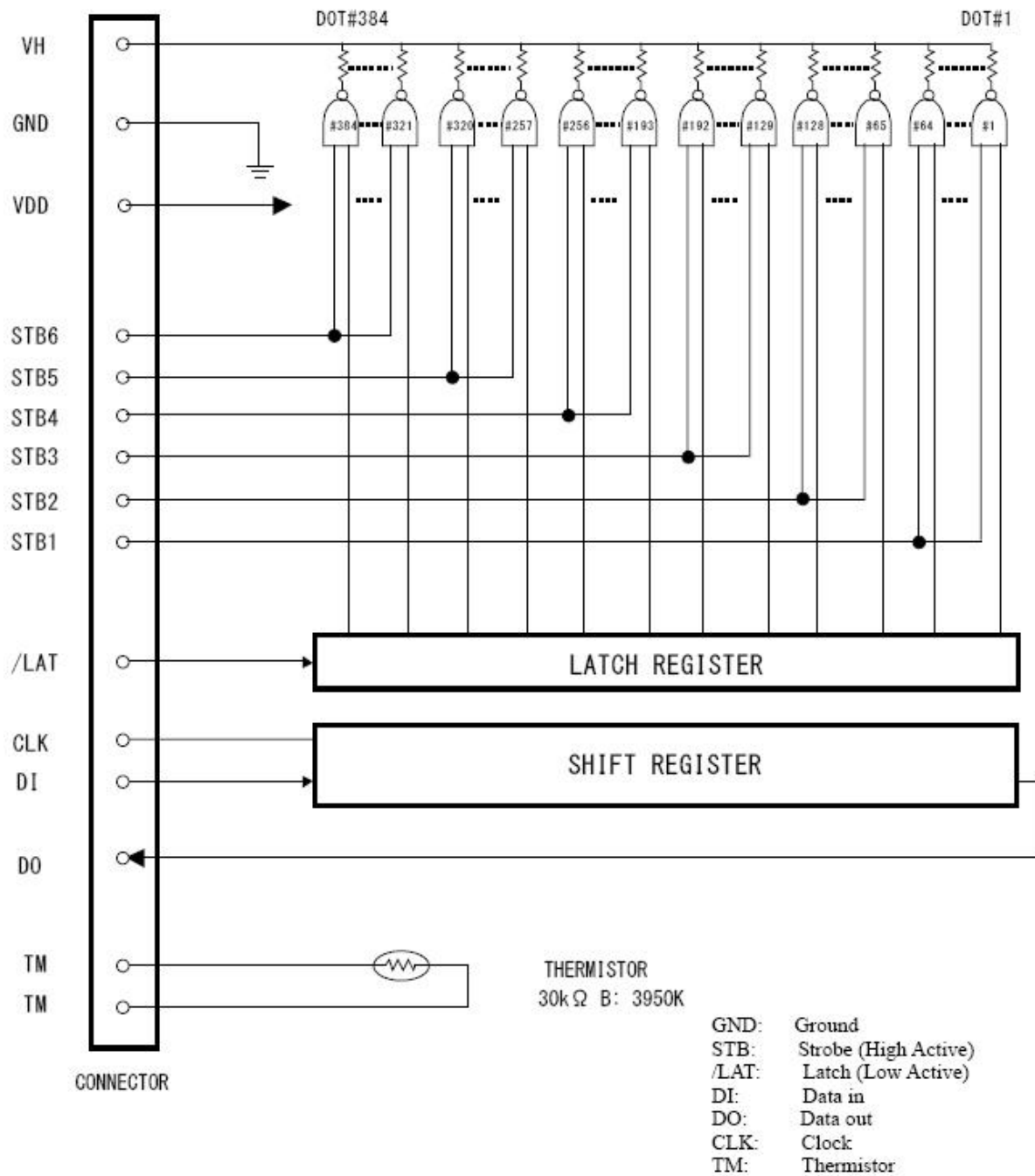
Item		Symbol	Min	Typical	Max	Unit	Condition
Print voltage		V _H			9.5	V	
Logic voltage		V _{DD}	2.7	3.0	3.9	V	
Logic current		I _{DD}			18	mA	f _{DI} =f _{CLK} /2 F _{clk} =6MHz
Input voltage	High	V _{IH}	0.8V _{DD}		V _{DD}	V	STB,DI,LAT,CLK
	Low	V _{IL}	0		0.2 V _{DD}	V	
Data input current	High	I _{IH} DI			0.5	μA	V _{IH} =3.3V V _{IL} =0V
	Low	I _{IL} DI			-0.5		
STB input current	High	I _{IH} STB			15		
	Low	I _{IL} STB			-0.5		
Clock input current	High	I _{IH} CLK			1.0		
	Low	I _{IL} CLK			-1.0		
Latch input current	High	I _{IH} LAT			1.0		
	Low	I _{IL} LAT			-1.0		
Output voltage	High	V _{DOH}	2.55			V	V _{DD} =3.0V, I _{OH} =-0.5mA
	Low	V _{DOL}			0.45	V	V _{DD} =3.0V, I _{OL} =0.5mA
Clock frequency		f _{CLK}			6	MHz	Review Fig.3
Clock width		tw _{CLK}	50			ns	
Data setup time		t _{setup} DI	70			ns	
Data retention time		thold DI	40			ns	
Data output delay time		td _{DO}			130	ns	
Latch pulse width		tw _{LAT}	100			ns	
Latch setup time		t _{setup} LAT	100			ns	
Latch retention time		thold LAT	40			ns	
STB setup time		t _{setup} STB	300			ns	
Heating to drive output delay		T _{do}			36	μs	

Fig.3 Timing Chart



*If delay time for Driver Out can not be secured enough, there is a possibility that VH would fluctuate greatly. Please design the circuit so that VH does not exceed peak voltage (V_p).

3.7 Structure picture



STB No.	Dot No.	dots/STB
1	1 ~ 64	64
2	65 ~ 128	64
3	129 ~ 192	64
4	193 ~ 256	64
5	257 ~ 320	64
6	321 ~ 384	64

3.8 Thermal resistance

$$R = R_{25} e^{B \left(\frac{1}{T+273} - \frac{1}{25+273} \right)}$$

Remark

R_{25} 30K Ω $\pm 5\%$

B 3950K $\pm 2\%$

T Temperature (°C)

Valid range -20°C ~ +80°C

Thermal resistance Temperature sheet:

T(°C)	Resistance (K Ω)	T(°C)	Resistance (K Ω)	T(°C)	Resistance (K Ω)	T(°C)	Resistance (K Ω)
-40	843	-10	161	20	37.5	50	10.8
-35	623	-5	124	25	30.0	55	8.91
-30	466	0	96.8	30	24.2	60	7.41
-25	352	5	75.7	35	19.6	65	6.2
-20	269	10	59.5	40	15.9	70	5.21
-15	208	15	47.1	45	13.1	75	4.4

3.9 Cautions

3.9.1 The TPH and photoelectric sensor on the movement are static sensitive devices. When using the movement, please take protective measures (such as wearing an electrostatic ring to ensure the humidity of the workshop) to prevent static damage to the internal components of the movement.。

3.9.2 When installing the rubber roller part to the bracket, be careful not to damage the rubber part of the rubber roller, the rubber roller gear and other bearing parts (especially, do not apply any oil or other foreign matter on the rubber part).

3.9.3 Do not touch the thermal head directly with your hand. When any thermal grease is stuck on the thermal head, the life of the thermal head will be greatly shortened. If the thermal head is stuck with any oil or foreign matter, wipe it with a cotton cloth dampened with alcohol. Allow the alcohol to dry before continuing to use. Also, do not hit or scratch the thermal head with a hard object.

3.9.4 The gold finger end of the FPC is not allowed to be directly touched by hand or tapped or scratched with a hard object. When the structure of the whole machine is designed, it should be in a relatively relaxed state in the spatial position, and should not be tightened and subjected to additional additional force; When the operator assembles, do not pull the FPC hard. When inserting and removing the FPC of the movement, be sure to carry out the power supply of the movement of the movement. The number of insertions and removals of the FPC connected to the drive board should not exceed 10 times. Please be sure to be parallel to the socket.

3.9.5 Do not bend the FPC as this may cause the FPC to be damaged or broken. If you want to bend the FPC, the camber must be greater than R1.

3.9.6 Since the movement is a paper-loading structure design, the rubber roller can be taken out by pulling the rubber roller part with a little force. If a paper jam occurs, pulling the paper too hard will cause the rubber roller gear to fall off or be damaged. Please do not pull the paper with force.

3.9.7 If the thermal head or the printing paper with water continues to be used, the thermal head will be electrolytically eroded and cause damage. Therefore, please pay attention to the following terms when using the movement.:

A: Turn off the printer's power when no use.

B: Please do not use damp paper.

C: Do not apply power if the humidity causes water condensation. If it does, immediately turn off the power. At the same time, let the thermal head dry before use. The movement is related to the environment (low temperature or high humidity), and the condensate may be due to evaporation of the paper used when the movement is printed at high speed. Therefore, please pay attention to the environment in which the movement is placed.

D: When condensation occurs, turn off the printer immediately and wait until it is completely dry before continuing.

3.9.8 If the movement is out of paper or not used for a long time, please be careful to separate the

thermal head from the rubber roller. If there is no paper during printing, stop the movement of the movement. If you print all the time without paper, it will cause damage to the printer core.

3.9.9 When printing continuously, the temperature of the thermal head circuit board (the thermistor detection temperature) should not exceed 65 °C to protect the internal IC of the thermal head; the motor surface temperature should not exceed 90 °C, in order to better protect the motor coil.

3.9.10 Keep the paper feed smooth when printing.

3.9.11 Use good quality thermal paper, because the thermal sensitivity of thermal paper has a great influence on the printing effect, and the paper with rough paper wears the print head seriously, which will shorten the life of the print head.

3.9.12 When donot printing, the print power must be turned off.

Heating control signal

When the printer core voltage is turned on/off, the heating control signal is turned off.

3.9.13 The following step is required to turn the printer on/off:

Turn on: 1) Vdd → 2) VH

Turn off: 1) VH → 2) Vdd

4 Step Motor Specifications

Every step of the stepper motor the paper feed pitch 0.0625 mm.

4.1 Step motor phase

TP-486F step motor use 2-2 phase drive, there are 4 positions. As shown in the following sheet:

POSITION	STEP1	STEP2	STEP3	STEP4
PA	-	-	+	+
PB	-	+	+	-
/PA	+	+	-	-
/PB	+	-	-	+

4.2 Step motor data

Item	specification	condition
Voltage	4.2~9.5V	Need to control motor heating
Step pitch	0.0625mm	
Phase resistance	$10\Omega \pm 7\%$	20℃
Phase current	0.357A	
Life	3000 hours	

4.3 Step motor driver

It is recommended to use a PWM method to drive a stepper motor, such as a driver chip such as the L3967. Different drive currents are used for different motor speeds. This can effectively reduce the heating of the stepping motor and effectively reduce the noise during the printing step. Under low voltage operation, Rohm company TPH 6846, Sanyo company 1836, 1838 are recommended.

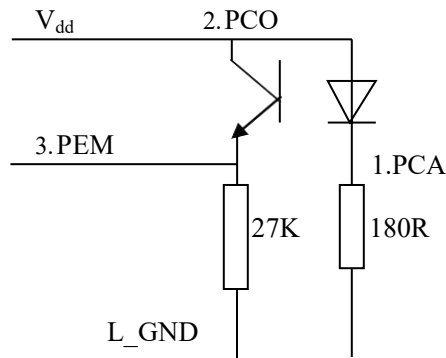
5 .Sensor specifications

TP-486F printer has a reflective photodetection switch. As shown in the figure below, when the paper is out of paper or the platen shaft is not pressed, the light from the photodetection cannot be reflected and output low level.

When both the paper and the platen shaft are normal, the light emitted by the photodetection is reflected and received by the receiving tube to output a high level.

The circuit drive of the photoelectric switch is shown in the figure below. The logic voltage can be used with 3.3V or 5V.

Do not activate printer heating when paper is out or the platen is not ready; when paper is out, the paper feed must be slowed down.

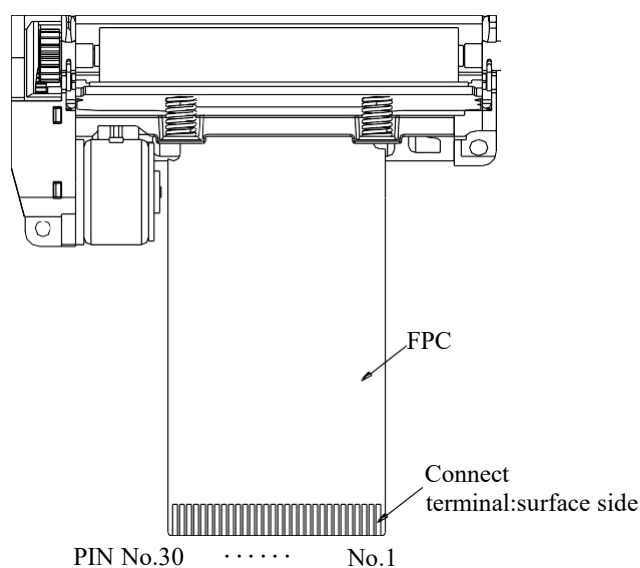


Optoelectronic data

Item		Symbol	condition	value			unit
Input	Forward voltage	V_F	$I_F=20\text{mA}$	1.0	1.2	1.5	V
	Reverse current	I_R	$V_R=5\text{V}$			10	μA
Output	From collector to emitter breakdown voltage	BV_{CEO}	$I_C=0.5\text{mA}$	30			V
	From emitter to collector Breakdown voltage	BV_{ECO}	$I_E=0.1\text{mA}$	5			V
	Collector current	I_{CEO}	$V_{CE}=10\text{V}$			100	nA
Coupling feature	Collector to emitter saturation voltage drop	$V_{CE(SAT)}$	$I_C=2\text{mA}$ $E_c=1\text{mW/cm}^2$			0.4	V
	Test distance	d					mm
	Leakage current	I_{LEAK}	$I_F=10\text{mA}$ $V_{CE}=5\text{V}$ No reflection surface			50	μA
	Rise/Fall time	t_r/t_f	$V_{CE}=5\text{V}$ $I_C=1\text{mA}$ $R_L=1000\Omega$			15/15	μs

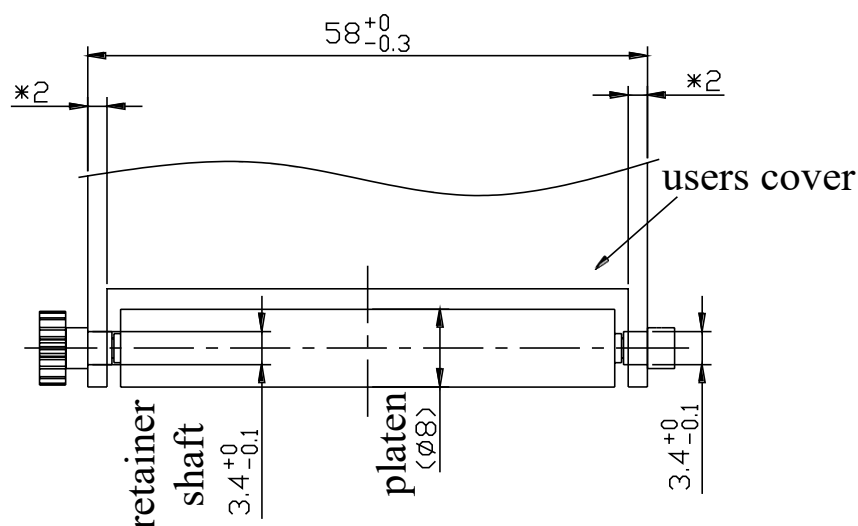
6. Connector (flexible) specifications

No	Symbol	Signal name
1	PCA	Power supply of the out-of-paper sensor (LED anode, photo-transistor collector)
2	PCO	Photo-transistor LED collector and emitter A
3	PEM	Output signal of the out-of-paper sensor (Photo-transistor emitter)
4~5	NC	No connection
6~7	VH	Thermal head drive power supply
8	DI	Print data input(serial input)
9	CLK	Synchronizing signal for print data transfer
10~11	P_GND	Head ground
12	STB6	Thermal head print activation instruction signal(#6 block)
13	STB5	Thermal head print activation instruction signal(#5 block)
14	STB4	Thermal head print activation instruction signal(#4 block)
15	Vdd	Logic power supply
16	TM	Thermistor
17	TM	Thermistor
18	STB3	Thermal head print activation instruction signal(#3 block)
19	STB2	Thermal head print activation instruction signal(#2 block)
20	STB1	Thermal head print activation instruction signal(#1 block)
21~22	P_GND	Head ground
23	\LAT	Print data latch (memory storage)
24	DO	Print data output(serial output)
25~26	VH	Thermal head drive power supply
27	PA	motor drive signal 1pin1
28	\PA	motor drive signal 1pin2
29	PB	motor drive signal 2pin1
30	\PB	motor drive signal 2pin2



7. Mechanical Design

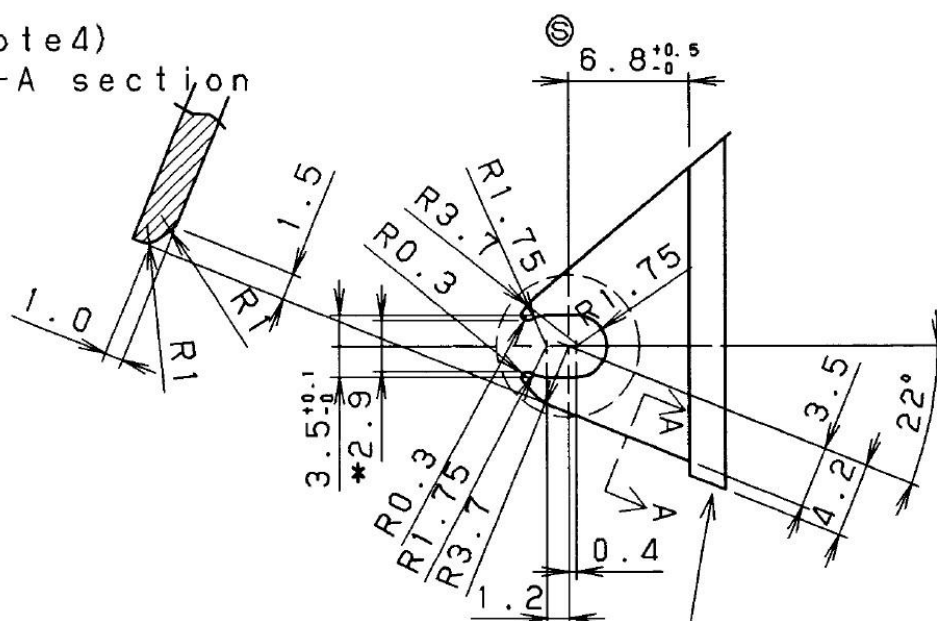
7.1 Easy loading paper design



Remark:

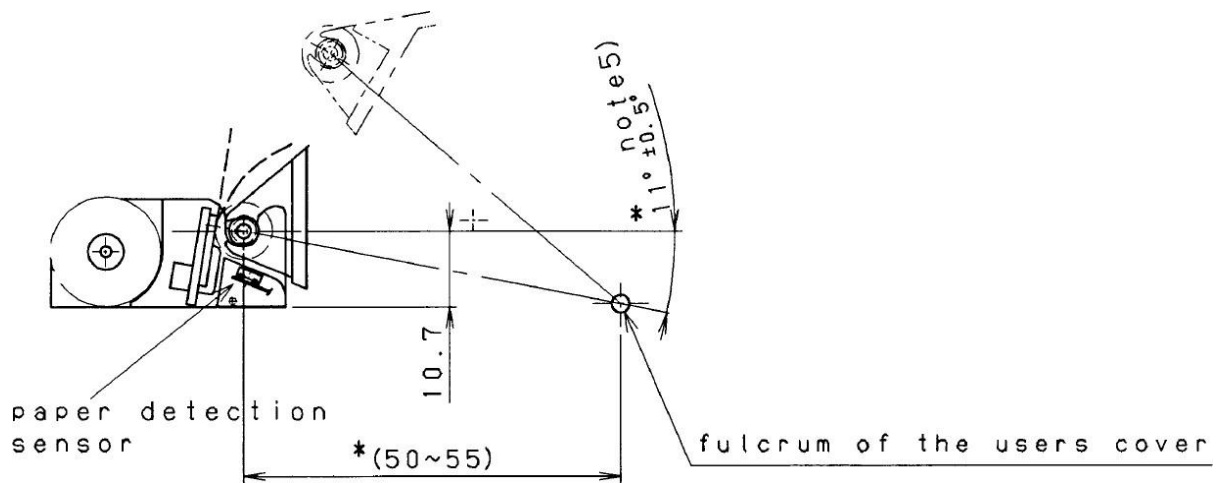
- 1) It is recommended size with “*” size
- 2) It is the center of printer of below ⑤
- 3) A paper guiding device is required to ensure the stability of the photoelectric detection signal output during paper feeding.
- 4) The sides of the easy to load paper structure design need to have a guide angle so that the rubber roller can be easily taken out from the movement and loaded when the paper is loaded.

note 4)
A-A section

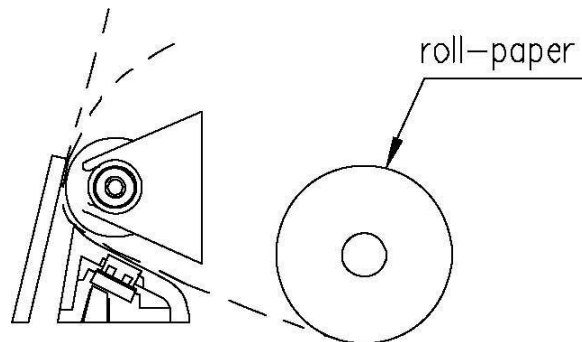


note 3)
paper holding guide (more than paper width)

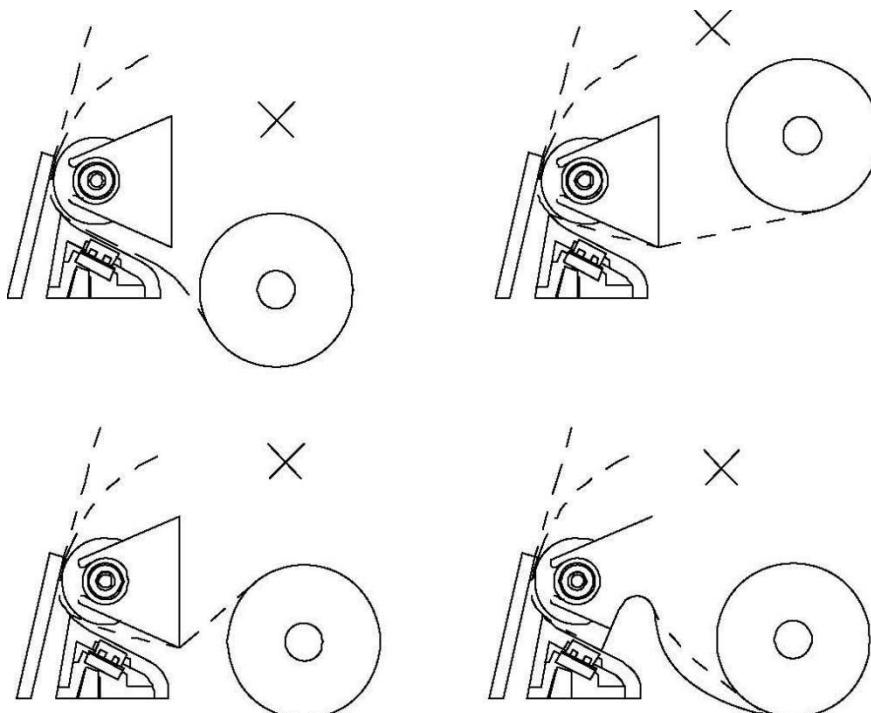
Easy to loading paper design position of the rotating shaft



7.2 Loading paper position



Wrong loading paper way:



7.4 Printer mechanism measure

